

QUBO Formulation: Benchmark Job Selection Problem

Benchmark Job Selection Problem

Problem Statement. Given a set of N candidate benchmark jobs, each characterized by a coverage reward v_i , a runtime t_i , and pairwise redundancy penalties p_{ij} , select a subset of jobs to maximize the total coverage reward while penalizing redundant coverage, subject to a total compute-time budget T .

Decision Variables. For each candidate job $i \in \{0, 1, \dots, N-1\}$, introduce a binary decision variable $x_i \in \{0, 1\}$, where $x_i = 1$ if job i is selected, and $x_i = 0$ otherwise.

QUBO Derivation. Step 1 — The original objective is a maximization problem. Standard QUBO solvers minimize energy, so we negate the objective:

$$\min_{x \in \{0,1\}^N} \left(- \sum_{i=0}^{N-1} v_i x_i + \sum_{0 \leq i < j \leq N-1} p_{ij} x_i x_j \right). \quad (1)$$

Step 2 — The compute-time budget constraint is expressed as:

$$\sum_{i=0}^{N-1} t_i x_i \leq T. \quad (2)$$

Step 3 — We convert the hard constraint into a quadratic penalty term. Following the standard QUBO convention, we add $\lambda \left(\sum_{i=0}^{N-1} t_i x_i - T \right)^2$ to the objective, where $\lambda > 0$ is a penalty weight. Expanding the square yields:

$$\lambda \left(\sum_{i=0}^{N-1} t_i^2 x_i^2 - 2T \sum_{i=0}^{N-1} t_i x_i + T^2 + \sum_{0 \leq i < j \leq N-1} 2t_i t_j x_i x_j \right). \quad (3)$$

Using the idempotent property of binary variables, $x_i^2 = x_i$, the expression simplifies to:

$$\lambda \sum_{i=0}^{N-1} (t_i^2 - 2T t_i) x_i + \lambda \sum_{0 \leq i < j \leq N-1} 2t_i t_j x_i x_j + \lambda T^2. \quad (4)$$

Step 4 — Combining the negated objective and the expanded penalty term gives the full QUBO Hamiltonian $H(x)$:

$$H(x) = \sum_{i=0}^{N-1} [-v_i + \lambda(t_i^2 - 2Tt_i)] x_i + \sum_{0 \leq i < j \leq N-1} [p_{ij} + 2\lambda t_i t_j] x_i x_j + \lambda T^2. \quad (5)$$

Step 5 — Reading off the Q matrix entries and the constant offset c from $H(x)$ in general closed form:

$$Q_{i,i} = -v_i + \lambda(t_i^2 - 2Tt_i), \quad i \in \{0, \dots, N-1\}, \quad (6)$$

$$Q_{i,j} = p_{ij} + 2\lambda t_i t_j, \quad 0 \leq i < j \leq N-1, \quad (7)$$

$$c = \lambda T^2. \quad (8)$$

Penalty Weight Justification. To guarantee that any budget violation is strictly penalized more than the maximum possible gain in the objective function, we require:

$$\lambda > \frac{\sum_{i=0}^{N-1} v_i}{(\min_i t_i)^2}. \quad (9)$$

This condition ensures that the quadratic penalty incurred by exceeding the budget dominates the total reward achievable by any feasible subset, thereby enforcing feasibility at the global minimum.

Correctness Argument. Minimizing $H(x)$ over $\{0, 1\}^N$ recovers the optimal solution because (i) all feasible assignments satisfy $\sum_i t_i x_i \leq T$, yielding a penalty term that evaluates to a constant offset c (or zero relative slack), and (ii) any infeasible assignment violates the budget by at least $\min_i t_i$, incurring a penalty of at least $\lambda(\min_i t_i)^2$. By the choice of λ , this penalty strictly exceeds the maximum possible reduction in the objective function from any feasible configuration. Consequently, the global minimum of $H(x)$ must correspond to a feasible assignment that optimally balances coverage rewards and redundancy penalties.

Illustrative Example. Consider a concrete instance with $N = 3$ jobs. Let the parameters be:

$$v_0 = 10, v_1 = 8, v_2 = 12; \quad t_0 = 3, t_1 = 4, t_2 = 5; \quad T = 7; \quad p_{01} = 2, p_{02} = 3, p_{12} = 1. \quad (10)$$

Choose $\lambda = 10$. Substituting these values into the general formulas yields the concrete Hamiltonian:

$$H(x) = -340x_0 - 408x_1 - 462x_2 + 242x_0x_1 + 303x_0x_2 + 401x_1x_2 + 490. \quad (11)$$

Evaluating $H(x)$ over all feasible subsets ($\sum_i t_i x_i \leq 7$):

$$H(\emptyset) = 490, \quad H(\{0\}) = 150, \quad H(\{1\}) = 82, \quad H(\{2\}) = 28, \quad H(\{0, 1\}) = -16. \quad (12)$$

The global minimum is $H(\{0, 1\}) = -16$, corresponding to the optimal selection of jobs 0 and 1, which yields a net coverage value of $v_0 + v_1 - p_{01} = 16$ while respecting the runtime budget $t_0 + t_1 = 7 \leq T$.